Chapter 13
Genomic Cross Prediction for Linseed Improvement

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Abstract Crossing between two or more parents is a fundamental way to generate superior genetic variants through genetic recombination and transgressive segregation in modern crop breeding. The selection of parents and crosses is the first key step for the success of crop breeding. The traditional method for screening parents and crosses is primarily based on phenotypic performance and genetic differences between parents and the breeders' empirical expertise. With the availability of genome-wide molecular markers and other genomic information, computer simulation offers a computational approach to simulate genetic recombination events between parents and progeny segregation of crosses and to generate segregation populations of any virtual crosses for various breeding schemes. Genomic selection (GS) enables to estimate breeding values (BVs) of the segregation individuals of crosses. Therefore, the integration of computer simulation and GS leads to an advanced genomic tool, named genomic cross prediction, to predict the genetic performance of different types of crosses by evaluating BVs and genetic variances of their segregation populations, enhancing the potential of success in crossbreeding. This chapter overviews the strategies and methods of genomic cross prediction and illustrates its application potential in crops, especially flax linseed breeding.

Keywords Flax · Linseed · Quantitative trait loci (QTL) · Genome-wide association studies (GWAS) · Genomic prediction · Molecular breeding · Cross prediction

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13.1 Introduction

Plant breeding by artificial selection has been exceptionally successful in developing new varieties that have contributed to the growth of modern societies and in satisfying the demand for plant-based products since the beginning of the domestication of plants, some 10,000 years ago (Fedoroff 2010). Phenotypic selection is the foundation of traditional breeding. Breeders select superior progenies to achieve genetic enhancement of target traits based on their expertise and the observed crop phenotypes. However, the success of this type of selection process depends on the genetic complexity of target traits. Most agronomic and economic traits are quantitative with complex genetic backgrounds and readily affected by environments; therefore, the evaluation of progeny populations of crosses needs to be performed in multiple years and locations. With the development of quantitative genetics and biostatistics, some statistical techniques have been used in plant breeding. For example, the best linear unbiased predictor (BLUP) that uses the progeny's phenotypic data and pedigree information was proposed to estimate breeding values (BVs) for assessing and selecting superior individuals (Henderson 1975). Since then, BLUP has been commonly used for genetic assessment in plant and animal breeding. Advances in molecular genetic methods have uncovered widespread genetic diversity in genomics since the 1990s. The vast numbers of molecular markers have been developed, allowing breeders to use markers to aid with breeding. Especially biparental populations-based quantitative trait locus (QTL) mapping has identified many large-effect QTLs. Marker-assisted selection (MAS) was then proposed (Lande and Thompson 1990) to improve the traits that are controlled by genes or QTLs with relatively large effects. As most of the economically important traits of crops are influenced by polygenes, each with a small effect (Riedelsheimer et al. 2012; Xu et al. 2012), the use of MAS in breeding practices is limited. Therefore, an improved method, genomic selection (GS), was introduced (Meuwissen et al. 2001). GS is considered an extension of MAS that uses genome-wide markers in a genotyped and phenotyped training population to build a statistical prediction model to predict BVs called genomic estimated breeding values (GEBVs) of unphenotyped individuals (Meuwissen et al. 2001). This landmark study laid the foundation for both plant and animal breeding to predict GEBVs of individuals and thus identify superior genotypes among selection candidates according to their genomic information.

Crossbreeding through crossing two or more parents is the fundamental method to generate superior genetic variants through genetic recombination between parents and their progeny's transgressive segregation in modern crop breeding. However, selecting parents to make crosses and predict the potential genetic performance of the crosses is the first critical step for the success of crossbreeding. This chapter introduces a new genomics-based strategy, named genomic cross prediction, which makes full use of information of genome-wide molecular markers and consensus genetic maps and integrates computer simulation and GS to simulate virtual crosses and predict the genetic performance of the virtual crosses to assist breeders in selecting parents and crosses in plant breeding effectively. A case study will be described to demonstrate the methods of genomic cross prediction in flax breeding.

13.2 Strategy of Genomic Cross Prediction

13.2.1 Genomic Cross Prediction

The fundamental objective of a breeding program is to develop superior cultivars for specific target traits under a wide range of environmental conditions. Crossbreeding using different types of crossing schemes such as single-, double-, or backcrosses, followed by progeny selection such as pedigree selection and single-seed descent, is the most commonly used method in plant breeding. Breeders routinely make many crosses every year and evaluate their progeny populations in the fields or the greenhouses. However, mostly very few of them outperform the check cultivars. Thus, parent evaluation and cross selection are critically important for crossbreeding. The traditional method for screening parents and crosses is usually based on parents' phenotypic performance and the difference between parents and breeders' empirical expertise. But the accuracy and efficiency of parent selection are impeded by unknown genetic structures, allelic makeup of the potential parents, and genetic performance of progeny populations. In practical breeding programs, making crosses using all potential genetic resources is impractical due to the extensive resources that would be required. However, the limited number of crosses will narrow the probability of finding the best recombinants.

Computer simulation is an efficient research tool used in various disciplines, including plant breeding, which can generate data that is unable or difficult to obtain from empirical experiments based on some theoretical considerations and/or empirical data. It is usually used to compare different methods and verify/validate proposed theoretical assumptions and models. Two types of computer simulation methods, deterministic and stochastic, are implemented in different simulation software tools for plant breeding studies. Deterministic simulation relies on genetic models derived from quantitative genetics theory with a set of parameter values and initial conditions, resulting in the deterministic output. DeltaGen (Jahufer and Luo 2018) is one of the software tools implementing the deterministic simulation and has been used to predict genetic gain and cost per selection cycle for different breeding strategies in forage species with empirical data. In contrast, stochastic simulation integrates some inherent variation and randomness of gene-to-phenotype relationship within the quantitative genetics framework (Hoyos-Villegas et al. 2019), leading to an ensemble of different outputs even with the same set of parameter values and initial conditions. Due to the nature of complex quantitative traits, stochastic simulation is more commonly used in plant breeding studies, for example, for strategic comparison of different breeding strategies (Wang et al. 2003,

2009) and evaluation of genomic selection in the breeding programs (Iwata and Jannink 2011; Lin et al. 2016; Sekine and Yabe 2020). With the availability of genome-wide molecular markers and the development of consensus genetic maps, computer simulation offers a computational approach to simulate genetic recombination between parents. It can generate segregation populations of numerous virtual crosses and generate segregation populations for crosses based on genome-wide markers on parents for various breeding schemes. On the other hand, GS provides an effective genomic approach to predict GEBVs of segregation populations, making it possible to evaluate the usefulness of crosses through the performance of their progeny populations.

GS has been studied in many crops, including flax, and emerged as one of the most promising breeding methods to improve genetic gains over conventional practices. GS has also been implemented in some practical breeding programs of crops, such as wheat (Crossa et al. 2013; Sun et al. 2020) and barley (Schmid and Thorwarth 2014), among others (Crossa et al. 2011). GS has demonstrated its potential for agronomic, abiotic, and biotic stress-related traits (You et al. 2016a; He et al. 2019a; Lan et al. 2020; Khan et al. 2021). The predictive ability of GS is largely based on the use of statistical models, the density of markers, and the relationship between training and testing populations (Desta and Ortiz 2014; Lipka et al. 2015). One of the fundamental features of GS is the use of high-density genome-wide markers. However, it usually results in low genomic predictive ability (GA) due to background noise and uncorrelated markers, along with possible high costs generated by genotyping of such a large number of markers in test populations (He et al. 2019a; Lan et al. 2020). The use of quantitative trait loci (QTLs) identified by genome-wide association studies (GWAS) can significantly improve predictive ability (He et al. 2019a; Lan et al. 2020). For instance, the predictive ability of pasmo resistance in flax was generated as high as 0.92 for 500 QTLs compared to 0.67 for 52,347 random SNPs (He et al. 2019a). Similarly, when QTLs were adopted in GS models, the predictive ability was 0.89 and 0.73 for seed yield (YLD) and days to maturity (DTM), respectively, in flax biparental populations. In contrast, it was 0.84 and 0.44 for YLD and DTM, respectively, when 17,277 genome-wide random SNPs were used (Lan et al. 2020). Similar results were also obtained for drought stress tolerance traits (Khan et al. 2021). Therefore, GS can be further fine-tuned by using a marker screening procedure to accelerate the rate of genetic gains in GS. The identification is one of the marker selection methods to select appropriate marker sets that have genetic correlation markers and breeding selection traits in the training population. The prediction ability of GS using QTL identified from the training population as markers relies on whether these QTLs also exist in the test population. Overall, the use of selected markers has the potential to enhances the predictive ability and reduces the number of markers likely to minimize genotyping costs, especially for selecting large breeding populations.

Therefore, the integration of computation simulation and GS offers an advanced genomic tool to predict the performance of different types of crosses, assisting breeders in making decisions in crosses to be made and increasing the potential of success in crossbreeding. Here we name this new genomic tool “genomic cross prediction.” The genomic cross prediction was initially proposed by Bernardo (2015) and applied to cross-evaluation of inbred lines in maize. In this pioneering study, a genetic resource panel of 284 diverse inbreds was phenotyped for several traits (flowering time, kernel composition, and disease resistance) and genotyped at 28,626 genome-wide SNPs in a previous study (Schaefer and Bernardo 2013). Then a similar procedure PopVar was described and used in barley (Mohammadi et al. 2015). This procedure has been implemented in the R package PopVar (Tiede et al. 2015a). A similar method was also used in wheat with single traits and a selection index for multiple traits (Yao et al. 2018). In this study, 57 wheat lines were used as a training population and genotyped with 7588 selected markers. The results showed that parental selection with the "usefulness" (definded in 13.2.3) resulted in higher genetic gain than midparent GEBVs. A selection index incorporating yield, extensibility, and maximum resistance as a new trait improved both yield and quality, while more genetic variance was retained in the selected progenies than the individual trait selection.

13.2.2 Procedure of Genomic Cross Prediction

We summarize the procedure of genomic cross prediction in Fig. 13.1. In this new procedure, the marker screening via GWAS has been integrated to improve the predictive ability of BVs of progenies using GS models. This procedure is described in detail as follows:

1. Genotyping and phenotyping of the genetic panel that includes genetic resources for potential parents. This panel is used as a training population to construct an optimal GS model for GEBV estimation of progenies.

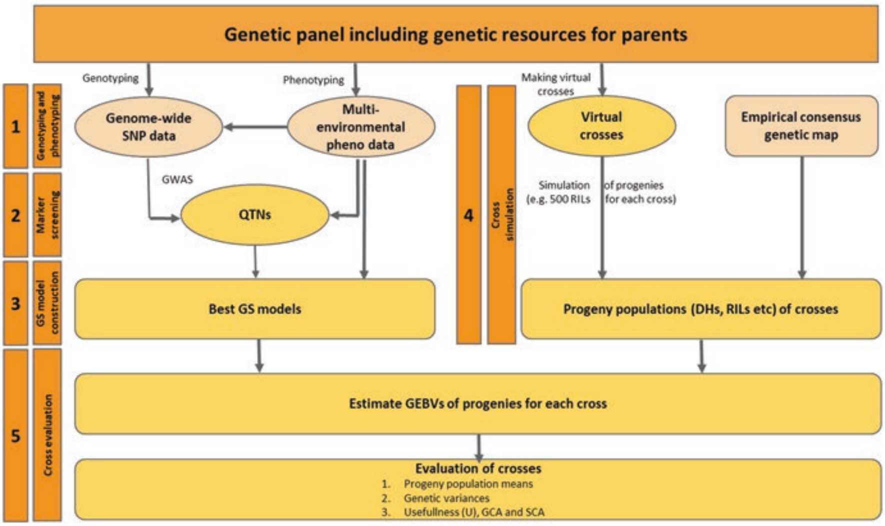

Fig. 13.1 A strategy of genomic cross prediction for crop improvement

2. Marker selection. Marker selection can be useful to reduce redundant markers from a large set of genome-wide random markers. The simultaneous use of several single- and multi-locus GWAS models to identify all potential large- and small-effect QTNs as markers (Lan et al. 2020) may be useful for genomic cross prediction in order to remove some of the SNPs unrelated to the traits of interest, especially in the case that the parents used to make crosses are selected from the training population for GS models.
3. GS model construction and cross-validation of models. Multiple GS models for a specific trait and a training population are compared and cross-validated to find an optimal GS model.
4. Cross simulation. Virtual crosses between all or a subset of parents in the genetic panel are made, and a certain number of progenies (e.g., 500) for each cross are simulated based on an empirical consensus genetic map; a specified crossing scheme such as single-, double-, or backcross; and a progeny advancing method, such as pedigree, single-seed descent (SSD) to generate doubled haploid (DH), or recombination inbreed line (RIL).
5. Cross-evaluation. The optimal GS models for the trait per se will be used to predict GEBVs of individuals in each progeny population. For each cross, some genetic parameters for the selection of parents and crosses will be calculated based on GEBVs of individuals. As a result, the potentially best parents and the top crosses will be selected to make crosses in practical breeding.

The prerequisites of genomic cross prediction include a panel of diverse genetic resources including potential parents and a consensus genetic map with high-density genome-wide markers. The genetic panel can be phenotyped over multiple years and locations and genotyped using different genotyping approaches, such as genotyping by sequencing. The accuracy of the genomic cross prediction depends mostly upon these two conditions. A large and diverse genetic panel as a training population helps to obtain a high predictive ability GS model. In particular, including the parents to be evaluated in the training population benefits the identification of QTLs shared by both the training population and the parents to be evaluated, increasing the genetic relationship between the training population and the progeny populations and thus promoting the predictive ability of progeny GEBVs. High-density genome-wide SNPs in the training population are also beneficial to the identification of all potential QTLs associated with the target traits and their genetic interactions (epistasis), which helps the simulation of true recombination events and the development of an optimal GS model. A high-density genetic map facilitates the estimation of genetic distances for identified QTLs. Ideally, the markers in the genetic map are a subset of the SNPs identified from the genetic panel.

13.2.3 Genetic Parameters for Cross-evaluation

According to the theories of quantitative genetics, a good cross should have a high progeny mean and large genetic variance that may have a high chance to obtain superior individuals from progenies. Thus, the performance for a cross can be measured by the progeny population mean and its genetic variance. Schnell and Utz (1975) defined a genetic parameter, usefulness (U) of a cross as

$$U = \mu \pm \Delta G = \mu \pm ih\sigma_g, \tag{13.1}$$

where μ is the mean of the progeny population derived from a cross, ΔG is a genetic gain, i is the standardized selection intensity based on the selected proportion (e.g., $i = 2.063$ for 5% or $i = 1.755$ for 10%), h is the square root of the trait heritability, and σ_g is the standard deviation of genetic variance of the progeny population. Zhong and Jannink (2007) further simplified this definition by setting $h = 1$ in the formula 13.1, resulting in

$$U = \mu \pm i\,\sigma_g \tag{13.2}$$

This new formula has a simpler property that expresses which crosses would generate progenies with higher genotypic values and thus is used in our procedure.

Midparent GEBVs of a cross can also be obtained from GS to evaluate whether midparent GEBV is adequate for predicting U.

13.3 Software Tools for Genomic Cross Prediction

13.3.1 Software Tools for Data Analysis

Genomic cross prediction involves both empirical and simulation data. Software tools are demanded to complete various genetic model development, data generation, and analysis. Based on the strategy of the genomic cross prediction diagrammed in Fig. 13.1, four categories of software tools are required: (1) GWAS to identify QTNs associated with the target traits or other marker screening methods, (2) GS model construction to find the optimal model to predict GEBVs of progeny populations, (3) genetic simulation to simulate virtual crosses and their progeny populations based on genomic data of parents, and (4) GEBV estimates of progeny populations and evaluation of cross performance. Table 13.1 lists some software tools that can be used for genomic cross prediction.

GWAS is usually used to identify QTNs related to the traits with a genetic panel that consists of potential genetic resources for parent selection. Two types of statistical models are available for QTL detection: single-locus models such as general linear model (GLM) (Price et al. 2006) and mixed linear model (MLM) (Yu et al. 2006) popularly used in the early QTL analyses and multi-locus models such as

Table 13.1 Software tools related to genomic cross prediction

Name	Description	Reference
GWAS		
mrMLM 4.0.2(R)	Implement six different multi-locus models, including mrMLM, FASTmrEMMA, ISIS EM-BLASSO, pLARmEB, pKWmEB, and FASTmrMLM	https://cran.r-project.org/web/packages/mrMLM/index.html
MVP 1.01(R)	Include single-locus models GLM and MLM and a multi-locus model FarmCPU	Liu et al. (2016)
Genomic selection		
rrBLUP 4.6.1(R)	Include a fast maximum-likelihood algorithm for mixed models	Endelman (2011)
BGLR 1.08(R)	Construct Bayesian regression models and GBLUP for continuous and categorical traits	Perez and de los Campos (2014)
Sommer 4.1.2(R)	GBLUP, rrBLUP	Covarrubias-Pazaran (2016)
Genetic simulation for breeding programs		
PedigreeSim 2.0 (R)	Simulate pedigreed populations for diploid and tetraploid species	Voorrips and Maliepaard (2012)
ADAM-plant (standard-alone Fortran program)	Simulate populations of various breeding schemes for both self- and cross-pollinated crops	Liu et al. (2019)
AlphaSim 0.13.0 (R)	Simulate plant and animal breeding programs	Faux et al. 2016
QuLine 2.5 (standard-alone)	Simulate breeding programs for cereal and leguminous crops, including male/female master selection, parent selection, single, backcross, top or double cross, and different progeny selection methods such as doubled haploid, marker-assisted selection etc.	Wang and Dieters (2008)
QuLinePlus0.0.10 (standard-alone)	Extension of QuLine for simulation of breeding populations of cross-pollinated crops	Hoyos-Villegas et al. (2019)
Blib	A generalized and powerful Fortran library to develop applications for various genetic modeling, simulation, and prediction in plant breeding	Personal communication with Dr. J Wang)
PopVar 1.3.0	Using phenotypic and genotypic data of a set of candidate parents to predict the mean, genetic variance, and superior progeny value of all or a subset of pairwise biparental crosses, and perform cross-validation to estimate genome-wide predictive ability of multiple statistical models	Tiede et al. (2015b)
MareyMap (1.3.6) (R)	A utility tool to calculate genetic distance between all markers on a physical map using a training genetic map	Siberchicot et al. (2017)

(R): R package

mrMLM (Wang et al. 2016; Li et al. 2017). The multi-locus models have a high statistical power to identify large and minor-effect QTNs that are genetic features of most complex quantitative traits. This provides us with a new option to directly use QTNs as markers in genomic cross prediction rather than genome-wide random markers to predict breeding values. The mrMLM package implements six different mixed models (mrMLM, FASTmrEMMA, ISIS EM-BLASSO, pLARmEB, pKW-mEB, and FASTmrMLM) (Wang et al. 2016; Li et al. 2017; Ren et al. 2017; Tamba et al. 2017; Zhang et al. 2017; Wen et al. 2018) (Table 13.1) that are complemental (He et al. 2019b; Lan et al. 2020). Thus, the combined results of these models are recommended to obtain as many associated markers as possible for GS model construction and cross simulation in genomic cross prediction.

Many genomic prediction models have been developed to increase genomic predictive ability, for example, ridge regression, best linear unbiased prediction (rrBLUP), genomic best linear unbiased prediction (GBLUP), Bayesian regression, partial least squares regression, and machine learning methods (Wang et al. 2018). These models may be roughly grouped into two groups based on the assumptions for statistical distributions of the marker effects. The models, such as RR-BLUP and GBLUP, assume that all markers contribute to the variation of the trait, while models such as BayesA and BayesB assume a specific variance for each marker. Thus, the first group of models is expected to be useful for complex quantitative traits with a polygenic architecture, while the second group of models is suitable for traits that are controlled by a small number of genes or QTL with large-effect architectures (e.g., Jannink et al. 2010; Spindel et al. 2015). Some studies have shown the better performance of BayesB than GBLUP or rrBLUP for traits controlled by a few genes with large effects (Daetwyler et al. 2010; Jannink et al. 2010; Thavamanikumar et al. 2015). However, if QTNs were used in the construction of GS models, similar predictive ability was obtained from different models, and rrBLUP or GBLUP is thus recommended because of their simplicity and computational efficiency (He et al. 2019a). Several R packages, including rrBLUP (Endelman 2011), BGLR (Perez and de los Campos 2014), and sommer (Covarrubias-Pazaran 2016), are available for GS model constructions (Table 13.1).

Making virtual crosses and simulating their progeny populations based on predefined breeding schemes are one of the steps for genomic cross prediction. Computer simulation has been widely used to simulate breeding schemes to save time and investigate problems that cannot be solved only by empirical data. Some software tools have been implemented to simulate various breeding schemes, such as single, double, and three crosses or backcrosses followed by different selection strategies for HD lines, RILs, etc. These tools include AlphaSim (Faux et al. 2016), pSBVB (Zingaretti et al. 2019), PedigreeSim (Voorrips and Maliepaard 2012), ADAM-plant (Liu et al. 2019), and QuLine or QuLinePlus (Wang and Dieters 2008). Notably, Blib developed by Dr. Jiankang Wang's laboratory (Chinese Academy of Agricultural Sciences, Beijing, China) is a universal library that has various functions to develop different applications of genetic model-ing, simulation, and prediction in plant breeding, including simulation of cross progeny populations.

Table 13.2 Components of a pipeline package of genomic cross prediction

Step	Module	Description
1	QTL mapping	Identify QTNs from the training population using a set of statistical models, especially multi-locus models
2	GSMoldeler	Construct and evaluate genomic prediction models using the training population that includes parents to be evaluated and the QTNs identified in Step 1 as markers
3	GeneticMapConversion	Generate a new genetic map covering all markers (QTNs) using a consensus genetic map as a training data set, using MareyMap (Siberchicot et al. 2017)
4	CrossSimulator	Simulate various virtual crosses and their genomic values of progeny populations using Blib
5	GEBVEstimator	Estimate GEBVs of progeny populations generated in Step 4 using GS models constructed in Step 2
6	CrossEvaluator	Evaluate performance of parents and crosses through analysis of progeny populations for the results in Step 5

13.3.2 A Pipeline Package of Genomic Cross Prediction

Although some third-party software tools are available for GWAS, GS modeling, and cross simulation, a pipeline program to integrate all these analyses is needed. PopVar is an earlier but practical pipeline program that includes GS modeling for multiple statistical models, cross simulation, and evaluation for a set of candidate parents and for all or a subset of pairwise biparental crosses (Table 13.1) (Mohammadi et al. 2015). We have developed a pipeline package that implements the strategy proposed in Fig. 13.1 and integrates all steps and eases the entire data analysis process. In particular, marker screening through GWAS and cross-evaluation by calculating genetic parameters of the cross are added to this pipeline. This pipeline package contains five program modules and is implemented in five separate pipeline programs (Table 13.2):

1. GWAS pipeline (a Perl program) that integrates all single-and multi-locus GWAS models implemented in R packages mrMLM and MVP (Step 1)
2. GS modeling and evaluation pipeline (a Java program) that integrates ten different GS models implemented in R packages rrBLUP, BGLR, and sommer (Step 2) and calculates GEBVs of simulated progeny populations (Step 5)
3. Genetic map conversion pipeline that integrates MareyMap (Siberchicot et al. 2017) to estimate the genetic distance of all markers used for genetic simulation based on a consensus genetic map as a training data set (a Perl program)
4. Cross simulation pipeline that combines the Blib library-based applications to simulate various crosses of breeding programs and generate genotypes of progeny populations (a Perl program) (Step 4)
5. Cross-evaluation pipeline to analyze various genetic parameters of crosses, including genetic means and variances, usefulness, etc. (Step 6)

13.4 Genomic Cross Prediction for Linseed Improvement

Pedigree analyses have shown that Canadian flax cultivars have a relatively narrow genetic base (You et al. 2016b). To broaden the narrow genetic base of Canadian flax cultivars, a core collection of 407 accessions has been selected from the world flax collections (Diederichsen et al. 2013; Soto-Cerda et al. 2013). These accessions originate from 39 countries in different ecological regions covering America, Europe, Asia, Oceania, and Africa and include some of the recently developed flax cultivars and superior breeding lines. They represent the majority of genetic variation in the world collection. This collection has been then fully phenotyped during 4 years and at 2 locations for more than 27 various traits, such as seed and fiber yield, seed and fiber quality, and disease resistance (You et al. 2017). A total of ~1.7 million SNPs have also been identified using a genotyping by sequencing (GBS) approach (He et al. 2019b). To date, crossbreeding is still a major breeding approach in flax breeding. To make full use of this phenotypic and genomic information in flax crossbreeding, as a case study, we applied the genomic cross prediction method to evaluate the performance of potential crosses and assist cross selection. Our idea was to use the well-phenotyped and well-genotyped core collection as a training population to develop optimal GS models, simulate progeny populations of all virtual crosses based on genomic data of parents, and then use the developed GS models to predict GEBVs of all progenies of crosses. Parent and cross performance were evaluated based on the estimates of general specific ability (GCA) of parents and the usefulness of the crosses.

13.4.1 Materials and Methods

13.4.1.1 Training Population and Phenotypic and Genomic Data

A total of 290 linseed accessions were extracted from the flax core collection as a training population. These accessions are potential parents in linseed breeding, including193 cultivars, 59 breeding lines, 13 landraces, and 25 unknown lines originating from 34 countries. A set of 258,708 SNPs in 290 linseed accessions extracted from the whole collection (He et al. 2019b) was extracted for analyses.

Five representative breeding target traits in linseed breeding selection, including seed yield (YLD), days to maturity (DTM), oil (OIL), linolenic acid (LIN), and powdery mildew resistance (PM), were chosen for the case study. The first four traits were phenotyped in 4 years (2009–2012) at two locations (Morden and Saskatoon, Canada), while PM was field evaluated for 5 years (2009–2013) in the PM nursery at Morden, Manitoba, Canada, which have been previously described in detail (You et al. 2017).

13.4.1.2 Identification of Quantitative Trait Nucleotides (QTNs)

QTNs of the five traits were detected using six multi-locus statistical models implemented in the R package mrMLM (Zhang et al. 2020), including mrMLM (Wang et al. 2016; Li et al. 2017), FASTmrMLM (Zhang and Tamba 2018), FASTmrEMMA (Wen et al. 2018), pLARmEB (Zhang et al. 2017), ISIS EM-BLASSO (Tamba et al. 2017), and pKWmEB. All individual phenotypic data sets from different years and locations were independently analyzed for GWAS, and then all detected nonredundant QTNs were combined for the downstream analyses. Significant QTNs were identified based on a cutoff value of the log of odds (LOD) score greater or equal to 3.0. The details of GWAS have been previously described (He et al. 2019b).

13.4.1.3 Construction of Genomic Selection Models

To select the optimal GS prediction models for different traits, ten GS models, including RR-BLUP, GBLUP, BayesA, BayesB, BayesC, BLL, BLR, RFR, RKHS, and SVR, were used to construct prediction models. The models were assessed using predictive ability with the fivefold cross-validation scheme (He et al. 2019a). Predictive ability was defined as a Pearson's correlation coefficient between predicted values and actual observed values. The optimal models were chosen to construct prediction models for traits using the data of all individuals in the training population to predict GBEVs of the crosses' progenies. To examine the impact of different marker sets on predictive ability, two different marker sets – all genome-wide random SNP markers and QTNs identified for each trait – were used to construct separate models.

13.4.1.4 Virtual Crosses and Simulation of Progeny Populations

All 290 linseed accessions were used to make possible virtual single crosses with a partial diallel cross scheme, i.e., a total of 41,905 (290 × 289/2) crosses were virtually made. For every single cross, 500 DH and 500 RIL individuals derived from two parents were simulated based on an additive model. Their genotypes were generated based on QTN markers in two parents and their genetic recombination in individuals. The consensus genetic map of flax (Cloutier et al. 2012) was used as a training data set to estimate genetic distance between neighboring markers for all QTNs. Since the consensus genetic map was SSR marker-based, we first anchored SSR markers to the flax scaffold sequences (Wang et al. 2012) and then mapped them to the pseudo molecule-scale flax reference sequence (You et al. 2018). Subsequently, the R package MereyMap (Chakravarti 1991; Siberchicot et al. 2017) was used to convert the physical distance of QTNs on chromosomes to genetic distance (cM). The optimal GS model for each trait was then used to predict the GEBVs of DH or RIL individuals for each cross and each trait.

13.4.1.5 Evaluation of Virtual Crosses

For each cross, GEBVs of all 500 progenies (DH or RIL individuals) were estimated using the GS model, and then its population mean (μ) and genetic variance (σ_g^2) were calculated. The usefulness (U) of a cross was calculated at the selected proportion of 5% based on formula 13.2: $U = \mu + i\ \sigma_g$ for YLD, OIL, and LIN to seek greater U values and $U = \mu\text{-}i\sigma_g$ for DTM and PM to seek smaller U values, where μ is the mean GEBVs of progenies, i is the standardized selection intensity 2.036, and σ_g is the standard deviation of genetic variance of the progenies.

GCA of a parent was defined as the average performance of this parent crossing with all other 289 parents, whereas U of a cross represents the specific performance of progeny populations by crossing two parents.

In the Canadian linseed breeding program, breeders select superior individuals of high seed yield and oil and linolenic content but short growth periods and high resistance to diseases (small ratings). Thus, selection for a parent or a cross is comprehensive for all major target traits, not only for a single trait. To comprehensively evaluate parents and crosses for all five traits, here we made an index trait that is a linear combination of all five traits with a specified weight for each trait, i.e., index trait $I = w1 \times x_{YLD} + w2 \times x_{DTM} + w3 \times x_{OIL} + w4 \times x_{LIN} + w5 \times x_{PM}$, where $w1$, $w2$, $w3$, $w4$, and $w5$ are the economic weights for YLD, DTM, OIL, LIN, and PM, respectively, with $w1 + w2 + w3 + w4 + w5 = 1$, and x is the GCA value of a parent or the U value of a cross for a single trait. Because the traits have different scales and units as well as different selection directions (high values for YLD, OIL, and LIN but low values for DTM and PM), the GCA values of parents or the U values of crosses were converted to relative values:

$x'_{Trait} = 1 - x_{Trait} / \max\left(x_{Trait}\right)$ for DTM and PM or

$x'_{Trait} = x_{Trait} / \max\left(x_{Trait}\right)$ for YLD, OIL, and LIN. As such, $I = w1 \times x'_{YLD} + w2 \times x'_{DTM} + w3 \times x'_{OIL} + w4 \times x'_{LIN} + w5 \times x'_{PM}$, with $0 \le I \le 1$. According to the relative importance of five traits in breeding, as an example, we assigned 0.4, 0.15, 0.15, 0.15, and 0.15 to $w1$, $w2$, $w3$, $w4$, and $w5$, respectively.

All computations for this study were performed using the pipeline package as described in 13.3.

13.4.2 Results and Discussions

13.4.2.1 Identification of Quantitative Trait Nucleotides (QTNs)

A total of 450, 317, 496, 313, and 235 nonredundant QTNs were identified using six multi-locus models for YLD, DTM, OIL, LIN, and PM, respectively, which had a range of large or minor QTN effects (Table 13.3). More QTNs were detected from YLD and OIL than from DTM, LIN, and PM. These outcomes of QTNs indicated varying genetic complexity or background for different traits.

Table 13.3 . Summary of phenotypic performance and quantitative trait nucleotides (QTNs) identified using multi-locus models for five traits from the training population of 290 linseed accessions.

Trait	Unit	Acr[a]	Mean ± s[b]	Min – Max	No. of QTNs	R^2 of QTNs (%)
Seed yield	t/ha	YLD	0.83 ± 0.28	0.17 – 1.36	450	1.13 – 38.30
Days to maturity	days	DTM	97.73 ± 3.64	89.06 – 109.71	317	0.63 – 17.66
Oil content	%	OIL	42.61 ± 1.71	37.74 – 50.69	496	0.80 – 23.57
Linolenic acid content	%	LIN	54.83 ± 5.30	5.02 – 66.07	313	0.10 – 17.86
Powdery mildew	0–9	PM	4.71 ± 1.26	2.50 – 8.00	235	0.72 – 14.40

[a]Acronym
[b]Means and standard deviation of traits over 4 years and two locations except for PM which is 5 years and one location

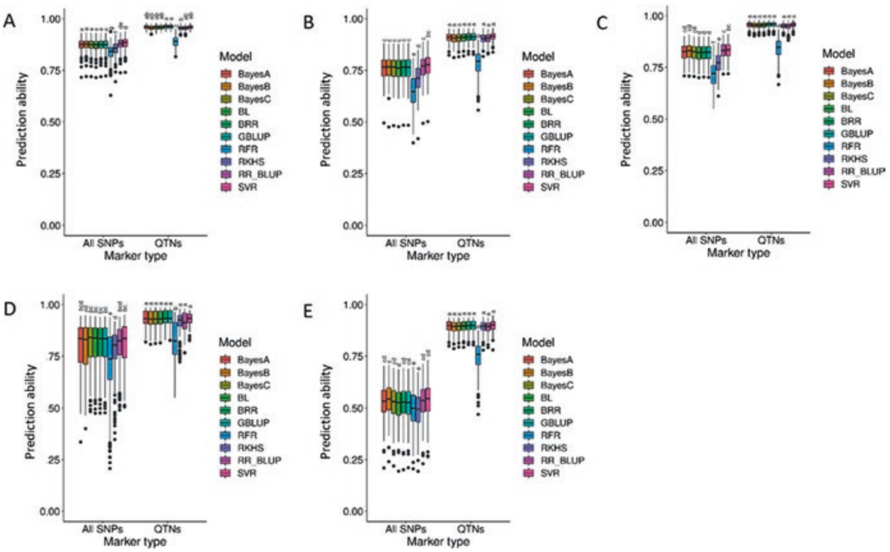

Fig. 13.2 Comparisons of predictive ability for ten genomic selection models with two different types of markers (all SNPs and QTNs). A fivefold cross-validation was used to estimate the predictive ability. Different letters represent statistical significance at a 5% probability level. (**a**). seed yield (t/ha); (**b**). days to maturity (days); (**c**). oil content (%); (**d**). linolenic acid content (%); (**e**). powdery mildew resistance

13.4.2.2 Optimal GS Models

To choose the optimal GS models, ten different GS models were compared using a fivefold cross-validation approach. Also, for each model, two types of markers, all genome-wide random markers (all SNPs) and QTNs identified for each trait, were used to construct models. The results were depicted in Fig. 13.2. Significant differences in predictive ability using two types of markers were observed for all five

Table 13.4 Predictive ability (*r*) using RR-BLUP obtained with the trait-specific QTNs and the All SNPs datasets for five traits using a five-fold cross-validation scheme. The population sizes in model evaluation are 217 and 72 for the training and test populations, respectively.

Trait	Unit	Acronym	*r* ± *s* for datasets	
			All SNPs	QTNs
Seed yield	t/ha	YLD	0.88 ± 0.03	0.95 ± 0.01
Days to maturity	days	DTM	0.76 ± 0.06	0.90 ± 0.02
Oil content	%	OIL	0.83 ± 0.04	0.95 ± 0.02
Linolenic acid content	%	LIN	0.80 ± 0.11	0.92 ± 0.04
Powdery mildew	0–9	PM	0.53 ± 0.08	0.89 ± 0.03

s standard deviation. For each trait, the predictive ability between All SNPs and QTNs is statistically significant at a 0.01 probability level

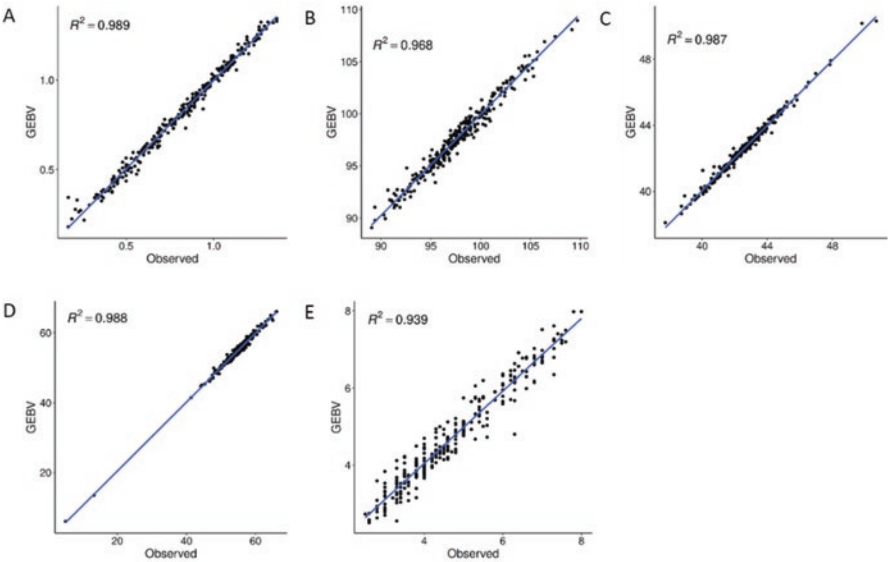

Fig. 13.3 Relationship between the general combining abilities (GCAs) and the GEBVs of parents of parents single crossed for five traits. (a). seed yield (t/ha); (b). days to maturity (days); (c). oil content (%); (d) (d). linolenic acid content (%); (e). powdery mildew resistance

traits. The GS models using QTNs generated consistently and significantly greater predictive ability than those using the genome-wide random markers (Table 13.4, Fig. 13.2). In particular, for PM, the predictive ability of the GS models using all SNPs was 0.53, compared to 0.89 when using the QTN dataset. For the QTN markers, except for RFR with significantly less predictive ability (0.75–0.89), all other nine GS models showed similarly high predictive ability (0.89–0.96) for all traits. For the genome-wide random SNP markers, ten models performed to be slightly different. Both RKHS and RFR had a low predictive ability, while the other eight models had no significant difference. Thus, in the following analyses, the RR-BLUP model with QTNs as markers was chosen to predict GEBVs of progeny populations of crosses because RR-BLUP has high computation efficiency. The RR-BLUP models constructed using respective QTNs for five traits generated very high model

R^2 (0.94–1.00) and prediction ability for themselves. The GEBVs of 290 accessions were highly correlated with the observed trait values for all five traits ($R^2 = 0.94$–0.99) (Fig. 13.3). The predictive abilities for all five traits were greater than 0.90, being 0.95, 0.90, 0.95, 0.92, and 0.89 for YLD, DTM, OIL, LIN, and PM, respectively (Table 13.4). Overall, the RR-BLUP models with QTNs of single trait per se as markers demonstrate high predictive ability and can be used to estimate GEBVs of progeny populations.

13.4.2.3 General Combining Ability (GCA) of Parents

In this study, we evaluated two progeny advancing methods, DH and RIL. The results show that the outcomes from both methods are highly similar ($R^2 = 0.993$–0.997) for all five traits. Here on end, only the results obtained from DH populations are displayed.

GCAs of 290 parents were calculated for all 41,905 single crosses. A consistently high linear relationship, i.e., close to 1, was observed between GCAs and GEBVs of the parents (Fig. 13.4), suggesting that GEBVs of parents estimated using GS models with QTN markers can effectively predict the GCAs of the parents.

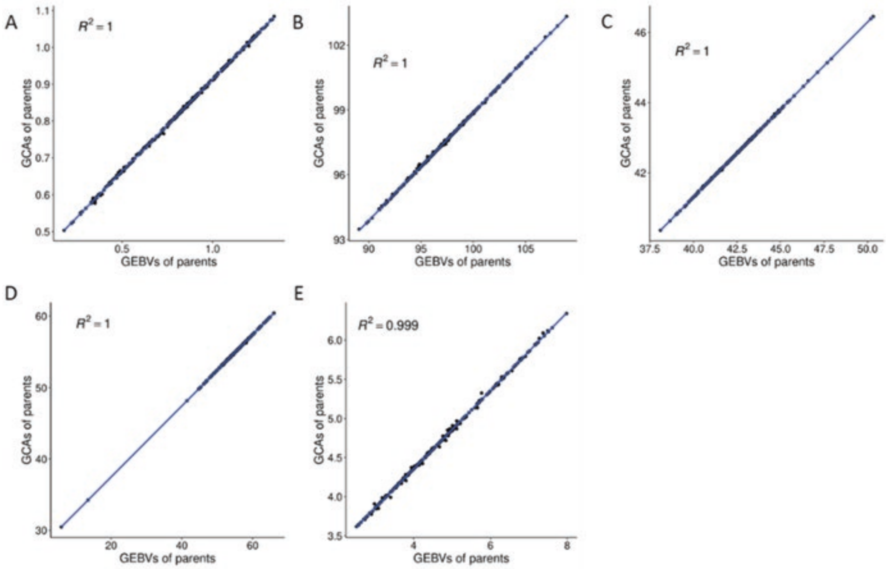

Fig. 13.4 Relationship between the general combining abilities (GCAs) and the GEBVs of parents of single crosses for five traits. (**a**). seed yield (t/ha); (**b**). days to maturity (days); (**c**). oil content (%); (**d**). linolenic acid content (%); (**e**). powdery mildew resistance

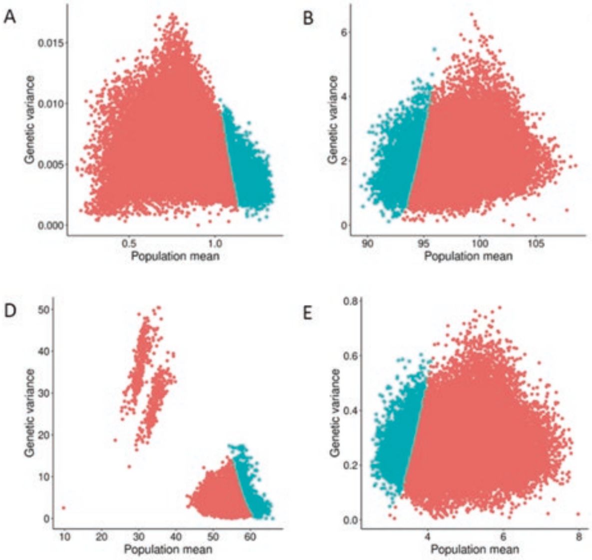

Fig. 13.5 Relationship between the population means and the genetic variances of progeny populations of single crosses for five traits. The blue dots represent the top 10% of crosses based on the usefulness *(U)* of the crosses at a 10% selection rate. (**a**). seed yield (t/ha); (**b**). days to maturity (days); (**c**). oil content (%); (**d**). linolenic acid content (%); (**e**). powdery mildew resistance

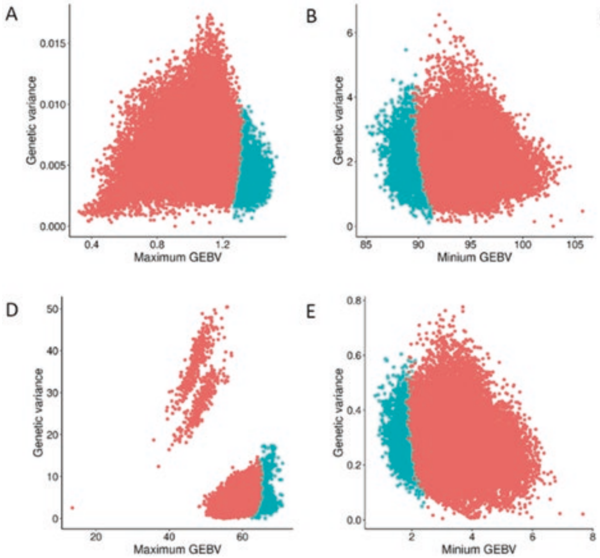

Fig. 13.6 Relationship between the minimum/maximum GEBVs and the genetic variance of progeny populations of single crosses for five traits. (**a**). seed yield (t/ha); (**b**). days to maturity (days); (**c**). oil content (%); (**d**). linolenic acid content (%); (**e**). powdery mildew resistance

13.4.2.4 Usefulness of Crosses

Us of 41,905 single crosses were calculated based on the 500 DH individuals simulated for each cross. *Us* were not linearly correlated with genetic variances of progeny populations. The top 10% of crosses (blue dots in Fig. 13.5) had high population means for YLD, OIL, and LIN or low population means for DTM and PM, but with the exception of OIL (Fig. 13.5c) they did not have the maximum genetic variation (Fig. 13.5a, b, d, and e). This was also true for the relationship between the minimum and/or maximum GEBV values and the genetic variance in the progeny populations the minimum or maximum values in prog-eny populations (Fig. 13.6). In linseed breeding programs, early maturing, disease resistant (minimum values) individuals with high YLD, OIL and LIN (maximum values) are selected.

13.4.2.5 Relationship of GCAs with *Us*

Significant linear relationship ($R^2 = 0.93$–0.98) between GCAs and *Us* was observed for all five traits (Fig. 13.7), showing that high GCA of a parent are high performant in crosses with other parents and results in superior crosses. Midparent value is often used to predict the performance of a cross. Fairly high correlations between midparent GEBVs and *Us* ($R^2 = 0.93$–0.98) were also observed (Fig. 13.8). The best

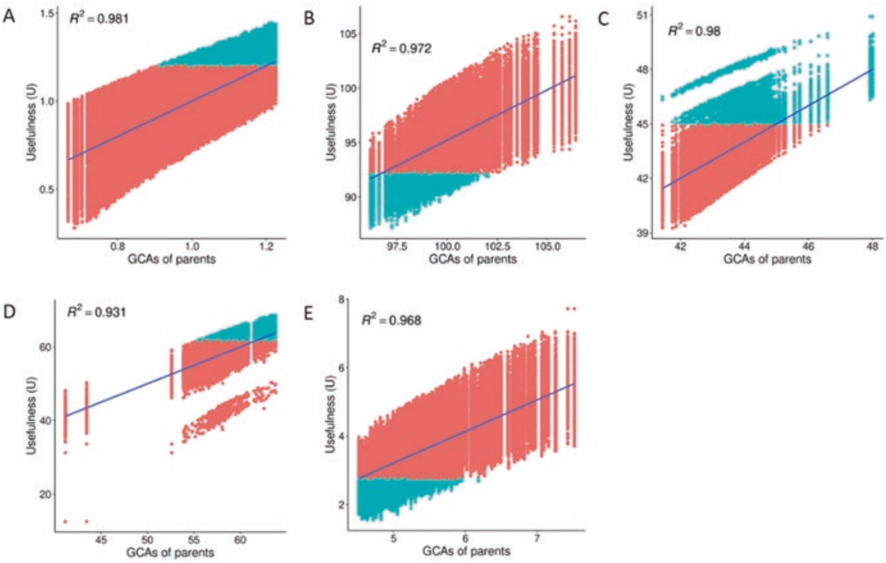

Fig. 13.7 Relationship between the general combining abilities (GCAs) of parents and the usefulness *(U)* of the corresponding crosses between parents for five traits. The blue dots represent the top 10% of crosses based on the usefulness of crosses at a 10% selection rate. (**a**). seed yield (t/ha); (**b**). days to maturity (days); (**c**). oil content (%); (**d**). linolenic acid content (%); (**e**). powdery mildew resistance

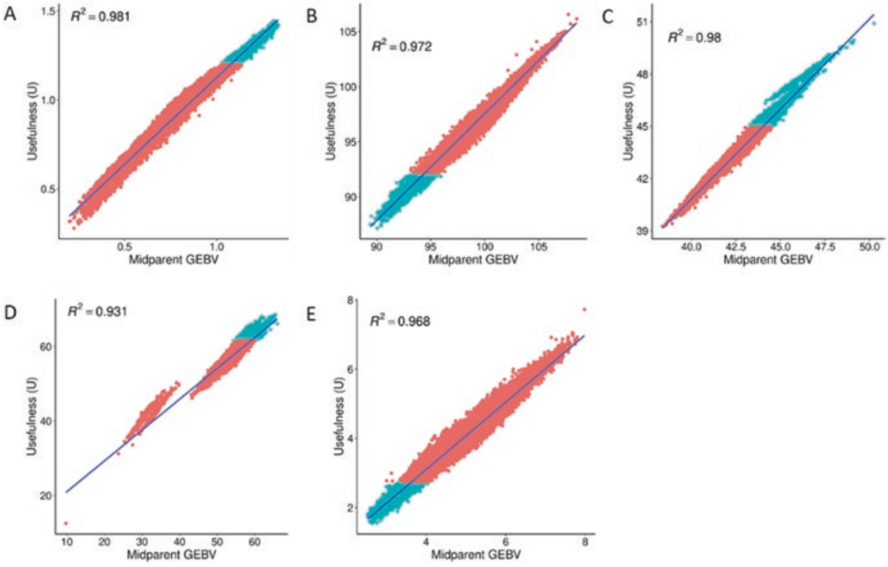

Fig. 13.8 Relationship between the midparent GEBV values and the usefulness *(U)* of single crosses for five traits. The blue dots represent the top 10% of crosses based on usefulness *(U)* of crosses at a 10% selection rate. (**a**). seed yield (t/ha); (**b**). days to maturity (days); (**c**). oil content (%); (**d**). linolenic acid content (%); (**e**). powdery mildew resistance

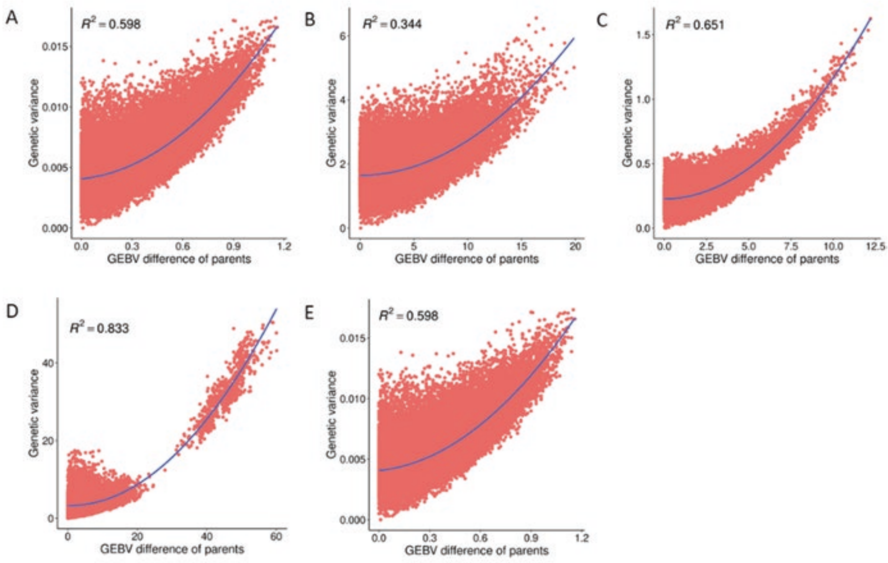

Fig. 13.9 Relationship between the GEBV differences of the two parents and the genetic variances of progeny populations of single crosses for five traits. The blue dots represent the top 10% of crosses based on usefulness *(U)* of crosses at a 10% selection rate. (**a**). seed yield (t/ha); (**b**). days to maturity (days); (**c**). oil content (%); (**d**). linolenic acid content (%); (**e**). powdery mildew resistance

crosses consistently had the best mid-parent GEBVs, confirming that it is indeed a good indicator of cross performance.

13.4.2.6 Differences Between Parents with Genetic Variance of Progeny Populations

The GEBV differences between the two parents reflected the genetic variation of progeny populations to some extent, varying in different traits (R^2 = 0.27–0.83) (Fig. 13.9). However, large parent differences between parents did not generate population mean (Fig. 13.5a, b, d and e). Once again, OIL performed differently from the other four traits, with the top crosses having a high genetic variance of populations, parent difference and parent GEBVs (Fig. 13.5c, 13.6c, and 13.9c). In particular for LIN, two low-LIN parents (Linola989 and CDC Gold) as parents with 5.02% and 13.26% LIN, respectively, resulting in large genetic variance in progeny populations (Fig. 13.9d), but did not generate superior crosses (superior population means) (Fig. 13.5d).

13.4.2.7 Evaluation of Top Parents and Crosses

Genomic cross prediction aims to predict superior parents of high GCAs and superior crosses of high *Us*. Table 13.5 lists the top 10% accessions (29 out of 290 accessions) with the highest GCAs for each trait, which have the best potential to improve traits in linseed crossbreeding. We observed that mostly unique subsets of high GCA accessions were obtained for different traits. However, 23 accessions were superior for two traits and the accession CN97907 (a USA cultivar) for three traits LIN, OIL, and PM. Using the 5-trait selection index, 25 out of the 29 parents were selected.

In the top 10% crosses (4090 out of 41,905 crosses), there were 51, 78, 212, 87, 60, and 214 parents involved in these crosses for YLD, DTM, OIL, LIN, PM, and the index trait, respectively. All 29 parents of the top 10% GCA were part of these crosses for all traits and the selection index trait (Table 13.6).

Table 13.5 Flax accessions of the top 10% general combining ability (GCA) values in the 290 potential linseed accessions for five traits and the selection index

Trait	Unit	Acronym	Accessions with top 10% general combining ability values		
Seed yield	t/ha	YLD	CAN_C_CDCMons	USA_C_CN97377	RUS_C_CN32542
			CAN_C_PrairieGrande	**USA_C_CN97873**	**CAN_C_CN19005**
			CAN_C_CN52732	ARG_C_CN97334	CAN_C_CN101413
			CAN_C_PrairieThunder	**CAN_C_CN19004**	CAN_C_CN33388
			USA_C_CN33992	UKR_U_CN101378	CAN_C_CN33385
			RUS_U_CN101348	USA_B_CN97670	CAN_C_CN18973
			USA_B_CN101286	**USA_C_CN18994**	RUS_C_CN97520
			CAN_C_CN18981	**USA_C_CN97444**	UNK_C_CN30861
			USA_B_CN100785	**ETH_B_CN19007**	**CAN_C_Shape**
			CAN_C_CN37286	**CAN_C_CN97392**	
Days to maturity	days	DTM	ARG_C_CN98014	RUS_L_CN97605	**IND_C_CN98157**
			DEU_C_CN97886	IND_C_CN98982	FRA_L_CN98742
			NLD_C_CN97616	FRA_C_CN97350	HUN_C_CN97300
			RUS_C_CN97484	ARG_C_CN97341	USA_B_CN98566B
			USA_C_CN97444	USA_B_CN97665	**CAN_C_CN97392**
			FRA_C_CN98794	USA_B_CN98566	**RUS_L_CN97483**
			IND_C_CN98135	CAN_C_CN97671	**RUS_C_CN97529**
			UNK_C_CN100547	IND_C_CN98364	**IND_U_CN98569**
			TUR_L_CN96958	TUR_U_CN101331	ARG_C_CN98027
			AFG_U_CN100952	**IND_C_CN97306**	

(continued)

Table 13.5 (continued)

Trait	Unit	Acronym	Accessions with top 10% general combining ability values		
Oil content	%	OIL	**RUS_B_ CN101137**	**RUS_B_ CN101307**	RUS_B_CN101279
			CAN_C_Shape	RUS_CN101402	PAK_C_CN97096
			CAN_B_ CN101596	PAK_C_ CN97092	RUS_B_CN101132
			IND_C_ CN98157	**IND_C_ CN97306**	USA_C_CN97396
			CAN_C_ CN19003	IND_L_ CN98242	FRA_B_CN98806
			USA_C_ CN97907	IND_L_ CN98240B	USA_C_CN98821
			CAN_C_ PrairieBlue	CAN_B_ CN101463	**IND_U_CN98569**
			IND_C_CN98363	**PAK_C_ CN97064**	**CAN_C_CN19005**
			FRA_C_ CN98734	FRA_C_ CN98807	FRA_B_CN98741
			IND_C_CN98250	**AFG_U_ CN101338**	
Linolenic acid content	%	LIN	**RUS_B_ CN101137**	**UNK_U_ CN100841**	CZE_C_CN100805
			RUS_B_ CN101307	NZL_B_ CN100797	FRA_C_CN98708
			CAN_B_ CN101463	NZL_B_ CN100797B	**CAN_B_CN101600**
			FRA_B_ CN100863	CAN_B_ CN101472	CAN_B_CN101454
			IND_C_ CN100799	ETH_C_ CN96988	SUN_C_CN100827
			RUS_C_ CN96846	**CHN_C_ CN101016**	USA_C_CN97393
			RUS_C_ CN96845	USA_C_ CN19160	ARG_C_CN97214
			CAN_C_ CN19004	**USA_C_ CN33992**	**FRA_C_CN18989**
			CAN_B_ CN101448	**USA_C_ CN97907**	CZE_C_CN98704
			CAN_C_ PrairieBlue	CAN_B_ CN101471	

(continued)

Table 13.5 (continued)

Trait	Unit	Acronym	Accessions with top 10% general combining ability values		
Powdery mildew	0–9	PM	RUS_C_CN97475	USA_C_CN33399	USA_C_CN98231
			USA_C_CN98541	**USA_C_CN97873**	TUR_C_CN96962
			ARG_C_CN97953	HUN_C_CN97287	TUR_U_CN100828
			GEO_U_CN101367	FRA_C_CN98773	**PAK_C_CN97064**
			IRN_L_CN97129B	FRA_L_CN98710	**UNK_U_CN100841**
			ETH_B_CN19007	**AFG_U_CN101338**	POL_B_CN98733
			MAR_C_CN98193	**USA_C_CN97907**	**CHN_C_CN101016**
			USA_B_CN97679B	TUR_U_CN100837	**RUS_C_CN97529**
			USA_C_CN18994	**FRA_C_CN18989**	USA_B_CN97679
			RUS_L_CN97483	**CAN_B_CN101600**	
Selection index			*ARG_C_CN97334*	*USA_C_CN97873*	*CAN_C_PrairieGrande*
			CAN_C_CDCMons	*CAN_C_PrairieThunder*	*CAN_C_CN19004*
			CAN_C_Shape	*USA_C_CN33992*	*USA_C_CN97377*
			USA_B_CN100785	*RUS_U_CN101348*	*CAN_C_CN18973*
			CAN_C_CN33385	*CAN_C_CN33388*	*CAN_C_CN101413*
			USA_C_CN18994	*UNK_C_CN30861*	*USA_B_CN97670*
			USA_C_CN97740	*USA_C_CN33399*	*CAN_C_CN19005*
			UKR_U_CN101378	*ETH_B_CN19007*	USA_C_CN97642
			RUS_L_CN97483	*CAN_C_CN37286*	*RUS_B_CN101132*
			CAN_C_CDCSorrel	TUR_U_CN101385	

Acron Acronym. Each accession name consists of three parts: country code (such as *CAN* Canada), development status (*C* cultivar, *L* landrace, *B* breeding line, *U* unknown), and CN numbers (unique accession identifier). For each trait, accession names are ordered by GCAs. The accessions that are in the top 10% subsets of at least two single traits are in bold type font, and the top 10% accessions for the selection index that are common to those for single traits are italicized.

s: standard deviation. Each accession name consists of three parts: country code (such as CAN: Canada), development status (C: cultivar; L: landrace; B: breeding line), and CN numbers (actual accession ID). For each trait, accession names are ordered by GCAs. The accessions that are in the top 10% subsets of at least two sing traits are bold, and the top 10% accessions for the index trait that are common to those for single traits are italicized.

Table 13.6 Best crosses for five traits and the selection index

Trait	Unit	Acronym	No. parents in top 10% crosses	Cross No.	Parent1	Parent2
Seed yield	t/ha	YLD	51	1	USA_C_CN97377	CAN_C_CDCMons
				2	USA_C_CN97873	CAN_C_PrairieGrande
				3	USA_C_CN97377	RUS_C_CN32542
				4	CAN_C_PrairieGrande	CAN_C_CDCMons
				5	USA_C_CN97873	USA_C_CN97377
				6	CAN_C_CN101413	CAN_C_CDCMons
				7	USA_C_CN97873	CAN_C_CN19005
				8	USA_C_CN97873	CAN_C_CN52732
				9	CAN_C_CDCMons	RUS_C_CN32542
				10	USA_C_CN97873	RUS_C_CN32542
Days to maturity	days	DTM	78	1	ARG_C_CN98014	RUS_L_CN97605
				2	IND_C_CN98157	RUS_L_CN97605
				3	ARG_C_CN98014	DEU_U_CN97886
				4	FRA_C_CN97350	RUS_L_CN97605
				5	ARG_C_CN98014	RUS_C_CN97484
				6	DEU_U_CN97886	IND_C_CN98157
				7	IND_C_CN98982	RUS_L_CN97605
				8	ARG_C_CN98014	IND_C_CN98982
				9	RUS_L_CN97605	FRA_L_CN98742
				10	ARG_C_CN98014	IND_C_CN98157

(continued)

Table 13.6 (continued)

Trait	Unit	Acronym	No. parents in top 10% crosses	Cross No.	Parent1	Parent2
Oil content	%	OIL	212	1	RUS_B_ CN101307	RUS_B_ CN101137
				2	RUS_B_ CN101279	RUS_B_ CN101137
				3	RUS_B_ CN101307	CAN_C_Shape
				4	RUS_B_ CN101137	CAN_C_Shape
				5	RUS_B_ CN101307	RUS_B_ CN101279
				6	RUS_B_ CN101137	CAN_B_ CN101596
				7	RUS_B_ CN101137	PAK_C_ CN97096
				8	RUS_B_ CN101307	RUS_U_ CN101402
				9	RUS_B_ CN101137	RUS_U_ CN101402
				10	RUS_B_ CN101307	PAK_C_ CN9709
Linolenic acid content	%	LIN	87	1	RUS_B_ CN101137	FRA_C_ CN98708
				2	FRA_C_CN98708	RUS_B_ CN101307
				3	NZL_B_ CN100797	FRA_C_ CN98708
				4	FRA_C_CN98708	CZE_C_ CN100805
				5	UNK_U_ CN100841	FRA_C_ CN98708
				6	FRA_C_CN98708	NZL_B_ CN100797B
				7	UNK_U_ CN100841	RUS_B_ CN101137
				8	FRA_C_CN98708	CAN_B_ CN101600
				9	CAN_B_ CN101463	FRA_C_ CN98708
				10	CZE_C_ CN100805	RUS_B_ CN101137

(continued)

Table 13.6 (continued)

Trait	Unit	Acronym	No. parents in top 10% crosses	Cross No.	Parent1	Parent2
Powdery mildew	0–9	PM	60	1	RUS_C_CN97475	USA_C_CN98541
				2	USA_C_CN98231	RUS_C_CN97475
				3	ARG_C_CN97953	USA_C_CN98541
				4	USA_C_CN33399	RUS_C_CN97475
				5	USA_C_CN98231	USA_C_CN98541
				6	USA_C_CN97873	PAK_C_CN97064
				7	HUN_C_CN97287	USA_C_CN98541
				8	TUR_C_CN96962	PAK_C_CN97064
				9	USA_C_CN97873	USA_C_CN98231
				10	USA_C_CN98541	FRA_L_CN98710
Selection index			214	1	ARG_C_CN97334	USA_C_CN97873
				2	USA_C_CN97377	USA_C_CN97873
				3	UNK_C_CN30861	USA_C_CN97873
				4	CAN_C_PrairieGrande	USA_C_CN97873
				5	USA_C_CN18994	USA_C_CN97873
				6	USA_C_CN33399	USA_C_CN97873
				7	USA_C_CN97873	USA_B_CN97670
				8	USA_C_CN97642	USA_C_CN97873
				9	CAN_C_Shape	USA_C_CN97873
				10	UKR_C_CN30860	USA_C_CN9787

13.5 Conclusions

With the availability of genome-wide molecular markers of potential breeding parents and other genomic information of biparental populations such as consensus genetic maps, a new genomics-based breeding tool, named genomic cross prediction, has been proposed to help breeders in assessing parents and choosing the potentially best crosses to make in breeding programs. The rationale of this genomic tool relies on the facts that the progeny populations of any number of virtual crosses can be easily simulated based on the genomic information of parents and the GS models built based on genome-wide QTLs identified from the parent panel can be effectively used to predict GEBVs of progenies. The evaluation of this genomic tool in maize, barley, wheat, and flax demonstrates its potential in breeding, offering an effective and low-cost breeding assisting tool. In particular, the case study in linseed demonstrates that the GEBVs of parents and midparent GEBVs are two good indicators for evaluating GCAs of parents and the usefulness of crosses, respectively. The comprehensive index traits combining several breeding target traits may benefit to the selection of parents and crosses with a superior comprehensive performance of several target traits.

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References

Bernardo R (2015) Genomewide selection of parental inbreds: classes of loci and virtual biparental populations. Crop Sci 55:2586–2595

Chakravarti A (1991) A graphical representation of genetic and physical maps: the Marey map. Genomics 11:219–222

Cloutier S, Ragupathy R, Miranda E, Radovanovic N, Reimer E et al (2012) Integrated consensus genetic and physical maps of flax (*Linum usitatissimum* L.). Theor Appl Genet 125:1783–1795

Covarrubias-Pazaran G (2016) Genome-assisted prediction of quantitative traits using the R package *sommer*. PLoS One 11:e0156744

Crossa J, Pérez P, de los Campos G, Mahuku G, Dreisigacker S, et al. (2011) Genomic selection and prediction in plant breeding. J Crop Imrov 25:239–261

Crossa J, Perez P, Hickey J, Burgueno J, Ornella L et al (2013) Genomic prediction in CIMMYT maize and wheat breeding programs. Heredity (Edinb) 112:48–60

Daetwyler, H.D.; Pong-Wong, R.; Villanueva, B.; Woolliams, J.A. (2010) The impact of genetic architecture on genome-wide evaluation methods. Genetics 185:1021–1031

Desta ZA, Ortiz R (2014) Genomic selection: genome-wide prediction in plant improvement. Trends Plant Sci 19:592–601

Diederichsen A, Kusters PM, Kessler D, Bainas Z, Gugel RK (2013) Assembling a core collection from the flax world collection maintained by plant gene resources of Canada. Genet Resour Crop Evol 60:1479–1485

Endelman JB (2011) Ridge regression and other kernels for genomic selection with R package rrBLUP. Plant Genome 4:250–255

Faux AM, Gorjanc G, Gaynor RC, Battagin M, Edwards SM et al (2016) AlphaSim: software for breeding program simulation. Plant Genome 9. https://doi.org/10.3835/plantgenome2016.3802.0013

Fedoroff NV (2010) The past, present and future of crop genetic modification. Nat Biotechnol 27:461–465

He L, Xiao J, Rashid KY, Jia G, Li P et al (2019a) Evaluation of genomic prediction for pasmo resistance in flax. Int J Mol Sci 20:359

He L, Xiao J, Rashid KY, Yao Z, Li P et al (2019b) Genome-wide association studies for pasmo resistance in flax (*Linum usitatissimum* L.). Front. Plant Sci 9:1982

Henderson CR (1975) Best linear unbiased estimation and prediction under a selection model. Biometrics 31:423–447

Hoyos-Villegas V, Arief VN, Yang W-H, Sun M, DeLacy IH et al (2019) QuLinePlus: extending plant breeding strategy and genetic model simulation to cross-pollinated populations—case studies in forage breeding. Heredity 122:684–695

Iwata H, Jannink JL (2011) Accuracy of genomic selection prediction in barley breeding programs: a simulation study based on the real single nucleotide polymorphism data ofbBarley breeding lines. Crop Sci 51:1915–1927

Jahufer MZZ, Luo D (2018) DeltaGen: a comprehensive decision support tool for plant breeders. Crop Sci. https://doi.org/10.2135/cropsci2017.2107.0456

Jannink, J.L. (2010) Dynamics of long-term genomic selection. Genet Sel Evol 42:35

Khan N, Shazadee H, Jia B, Bartaula S, Sertse D et al (2021) Genomic prediction for drought stress and root development traits in flax (*Linum usitatissimum* L.). Front Plant Sci

Lan S, Zheng C, Hauck K, McCausland M, Duguid SD et al (2020) Genomic prediction accuracy of seven breeding selection traits improved by QTL identification in flax. Int J Mol Sci 21:1577

Lande R, Thompson R (1990) Efficiency of marker-assisted selection in the improvement of quantitative traits. Genetics 124:743–756

Li H, Zhang L, Hu J, Zhang F, Chen B et al (2017) Genome-wide association mapping reveals the genetic control underlying branch angle in rapeseed (*Brassica napus* L.). front. Plant Sci 8:1054

Lin Z, Cogan NO, Pembleton LW, Spangenberg GC, Forster JW et al (2016) Genetic gain and inbreeding from genomic selection in a simulated commercial breeding program for perennial ryegrass. Plant Genome 9. https://doi.org/10.3835/plantgenome2015.3806.0046

Lipka AE, Kandianis CB, Hudson ME, Yu J, Drnevich J et al (2015) From association to prediction: statistical methods for the dissection and selection of complex traits in plants. Curr Opin Plant Biol 24:110–118

Liu X, Huang M, Fan B, Buckler ES, Zhang Z (2016) Iterative usage of fixed and random effect models for powerful and efficient genome-wide association studies. PLoS Genet 12:e1005767

Liu H, Tessema BB, Jensen J, Cericola F, Andersen JR et al (2019) ADAM-plant: a software for stochastic simulations of plant breeding from molecular to phenotypic level and from simple selection to complex speed breeding programs. Front Plant Sci 9:1926

Meuwissen THE, Hayes BJ, Goddard ME (2001) Prediction of total genetic value using genome-wide dense marker maps. Genetics 157:1819–1829

Mohammadi M, Tiede T, Smith KP (2015) PopVar: a genome-wide procedure for predicting genetic variance and correlated response in biparental breeding populations. Crop Sci 55:2068–2077

Perez P, de los Campos G (2014) Genome-wide regression and prediction with the BGLR statistical package. Genetics 198:483–495

Price AL, Patterson NJ, Plenge RM, Weinblatt ME, Shadick NA et al (2006) Principal components analysis corrects for stratification in genome-wide association studies. Nat Genet 38:904–909

Ren WL, Wen YJ, Dunwell JM, Zhang YM (2017) pKWmEB: integration of Kruskal-Wallis test with empirical Bayes under polygenic background control for multi-locus genome-wide association study. Heredity (Edinb) 120:208–218

Riedelsheimer C, Czedik-Eysenberg A, Grieder C, Lisec J, Technow F et al (2012) Genomic and metabolic prediction of complex heterotic traits in hybrid maize. Nat Genet 44:217–220

Schaefer CM, Bernardo R (2013) Genomewide association mapping of flowering time, kernel composition, and disease resistance in historical Minnesota maize inbreds. Crop Sci 53:2518–2529

Schmid KJ, Thorwarth P (2014) Genomic selection in barley breeding. In: Kumlehn J, Stein N (eds) Biotechnological approaches to barley improvement biotechnology in agriculture and forestry. Springer, Berlin/Heidelberg, pp 367–378

Schnell FW, Utz HF (1975) F1-leistung und elternwahl euphy-derzu ¨chtungvonselbstbefruchtern,. in Bericht u ¨ber die Arbeitstagung der Vereinigung O "sterreichischer Pflanzenzu" chter BAL Gumpenstein, Gumpenstein, Austria:243–248

Sekine D, Yabe S (2020) Simulation-based optimization of genomic selection scheme for accelerating genetic gain while preventing inbreeding depression in onion breeding. Breed Sci 70:594–604

Siberchicot A, Bessy A, Gueguen L, Marais GAB (2017) MareyMap online: a user-friendly web application and database service for estimating recombination rates using physical and genetic maps. Genome Biol Evol 9:2506–2509

Soto-Cerda BJ, Diederichsen A, Ragupathy R, Cloutier S (2013) Genetic characterization of a core collection of flax (*Linum usitatissimum* L.) suitable for association mapping studies and evidence of divergent selection between fiber and linseed types. BMC Plant Biol 13:78

Spindel, J.; Begum, H.; Akdemir, D.; Virk, P.; Collard, B.; Redona, E.; Atlin, G.; Jannink, J.L.; McCouch, S.R. (2015) Genomic selection and association mapping in rice (*Oryza sativa*): effect of trait genetic architecture, training population composition, marker number and statistical model on accuracy of rice genomic selection in elite, tropical rice breeding lines. PLoS Genet. 2015, 11, (2), e1004982.

Sun J, Khan M, Amir R, Gul A (2020) Genomic selection in wheat breeding. In: Ozturk M, Gul A (eds) Climate change and food security with emphasis on wheat. Academic Press, pp 321–330

Tamba CL, Ni YL, Zhang YM (2017) Iterative sure independence screening EM-Bayesian LASSO algorithm for multi-locus genome-wide association studies. PLoS Comput Biol 13:e1005357

Thavamanikumar S, Dolferus R, Thumma BR (2015) Comparison of genomic selection models to predict flowering time and spike grain number in two hexaploid wheat doubled haploid populations. G3 (Bethesda) 5:1991–1998

Tiede T, Kumar L, Mohammadi M, Smith KP (2015a) Predicting genetic variance in bi-parental breeding populations is more accurate when explicitly modeling the segregation of informative genomewide markers. Mol Breed 35:199

Tiede T, Mohammadi M, Smith KP (2015b) PopVar: genomic breeding tools: genetic variance prediction and cross-validation. R package version 121

Voorrips RE, Maliepaard CA (2012) The simulation of meiosis in diploid and tetraploid organisms using various genetic models. BMC Bioinform 13:248

Wang J, Dieters D (2008) QuLine, A software that simulates breeding programs for developing inbred lines Version 2.1. User's Manual

Wang J, van Ginkel M, Podlich D, Ye G, Trethowan R et al (2003) Comparison of two breeding strategies by computer simulation. Crop Sci 43:1764–1773

Wang J, Singh RP, Braun HJ, Pfeiffer WH (2009) Investigating the efficiency of the single backcrossing breeding strategy through computer simulation. Theor Appl Genet 118:683–694

Wang Z, Hobson N, Galindo L, Zhu S, Shi D et al (2012) The genome of flax (*Linum usitatissimum*) assembled de novo from short shotgun sequence reads. Plant J 72:461–473

Wang SB, Feng JY, Ren WL, Huang B, Zhou L et al (2016) Improving power and accuracy of genome-wide association studies via a multi-locus mixed linear model methodology. Sci Rep 6:19444

Wang X, Xu Y, Hu ZL, Xu CW (2018) Genomic selection methods for crop improvement: current status and prospects. Crop J 6:330–340

Wen YJ, Zhang H, Ni YL, Huang B, Zhang J et al (2018) Methodological implementation of mixed linear models in multi-locus genome-wide association studies. Brief Bioinform 19:700–712

Xu Y, Lu Y, Xie C, Gao S, Wan J et al (2012) Whole-genome strategies for marker-assisted plant breeding. Mol Breed 29:833–854

Yao J, Zhao D, Chen X, Zhang Y, Wang J (2018) Use of genomic selection and breeding simulation in cross prediction for improvement of yield and quality in wheat (*Triticum aestivum* L.). Crop J 6:353–365

You FM, Booker HM, Duguid SD, Jia G, Cloutier S (2016a) Accuracy of genomic selection in biparental populations of flax (*Linum usitatissimum* L.). Crop J 4:290–303

You FM, Duguid SD, Lam I, Cloutier S, Rashid KY et al (2016b) Pedigrees and genetic base of flax cultivars registered in Canada. Can J Plant Sci 96:837–852

You FM, Jia G, Xiao J, Duguid SD, Rashid KY et al (2017) Genetic variability of 27 traits in a core collection of flax (*Linum usitatissimum* L.). front. Plant Sci 8:1636

You FM, Xiao J, Li P, Yao Z, Gao J et al (2018) Chromosome-scale pseudomolecules refined by optical, physical, and genetic maps in flax. Plant J 95:371–384

Yu J, Pressoir G, Briggs WH, Vroh Bi I, Yamasaki M et al (2006) A unified mixed-model method for association mapping that accounts for multiple levels of relatedness. Nat Genet 38:203–208

Zhang YM, Tamba CL (2018) A fast mrMLM algorithm for multi-locus genome-wide association studies. bioRxiv. https://doi.org/10.1101/341784

Zhang J, Feng JY, Ni YL, Wen YJ, Niu Y et al (2017) pLARmEB: integration of least angle regression with empirical Bayes for multilocus genome-wide association studies. Heredity (Edinb) 118:517–524

Zhang Y-W, Tamba CL, Wen Y-J, Li P, Ren W-L et al (2020) mrMLM v4.0: an R platform for multi-locus genome-wide association studies. BioRxiv. https://doi.org/10.1101/2020.1103.1104.976464

Zhong, S.; Jannink, J.-L. (2007) Using quantitative trait loci results to discriminate among crosses on the basis of their progeny mean and variance. Genetics 177, 567–576

Zingaretti ML, Monfort A, Perez-Enciso M (2019) pSBVB: a versatile simulation tool to evaluate genomic selection in polyploid species. G3 (Bethesda) 9:327–334